

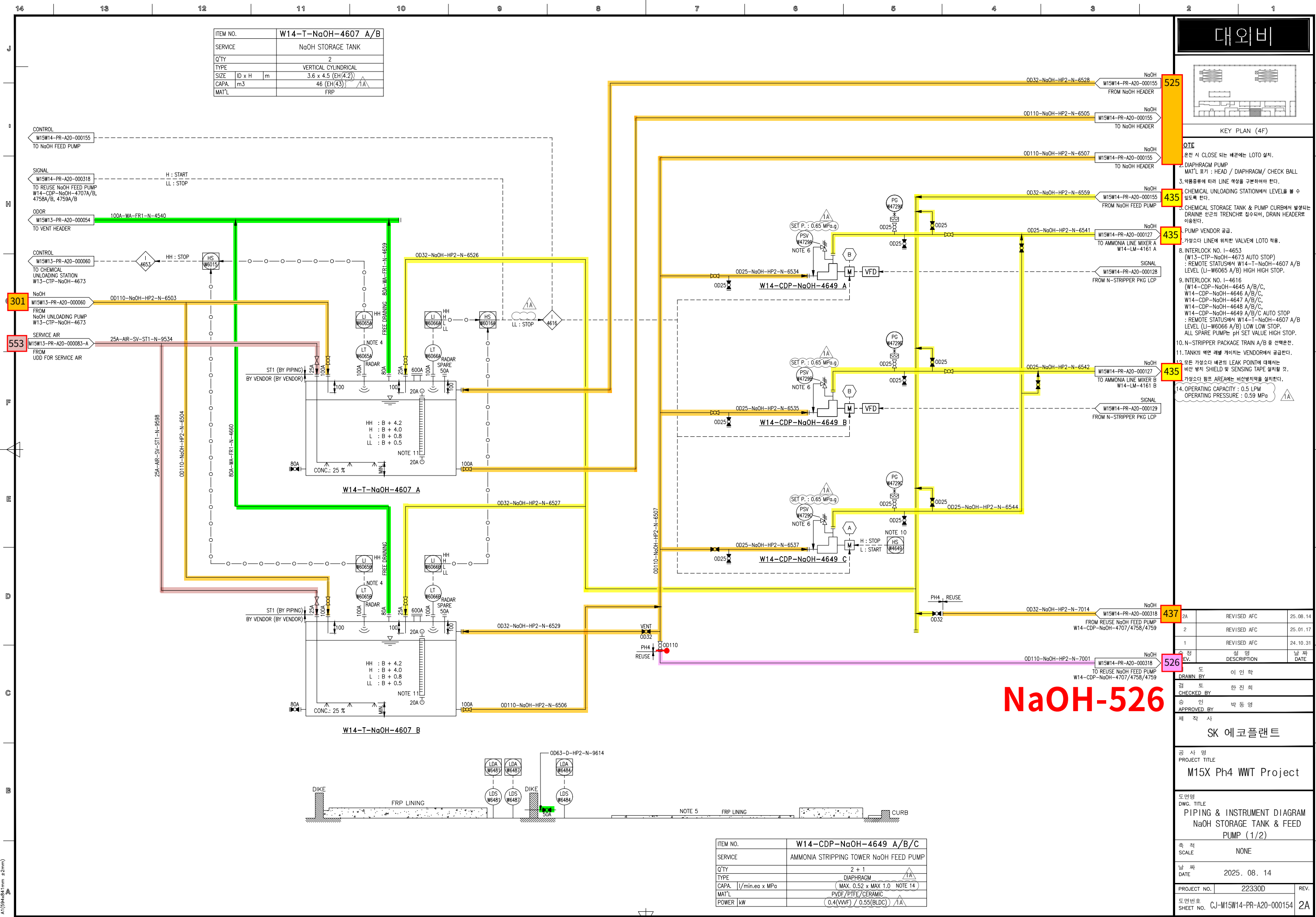
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

NaOH-526

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	526	4FA	OD25-NaOH-HP2-N-7002-01	1 OF 1	0	CJ-M15W14-PR-A20-000318	1A	
2	526	4FA	OD25-NaOH-HP2-N-7003-01	1 OF 1	0	CJ-M15W14-PR-A20-000318	1A	
3	526	4FA	OD25-NaOH-HP2-N-7006-01	1 OF 1	0	CJ-M15W14-PR-A20-000318	1A	
4	526	4FA	OD25-NaOH-HP2-N-7007-01	1 OF 1	0	CJ-M15W14-PR-A20-000318	1A	
5	526	4FA	OD32-NaOH-HP2-N-7004-01	1 OF 1	0	CJ-M15W14-PR-A20-000318	1A	
6	526	4FA	OD32-NaOH-HP2-N-7005-01	1 OF 1	0	CJ-M15W14-PR-A20-000318	1A	
7	526	4FA	OD110-NaOH-HP2-N-7001-01	1 OF 1	0	CJ-M15W14-PR-A20-000154, 318	2A/1A	



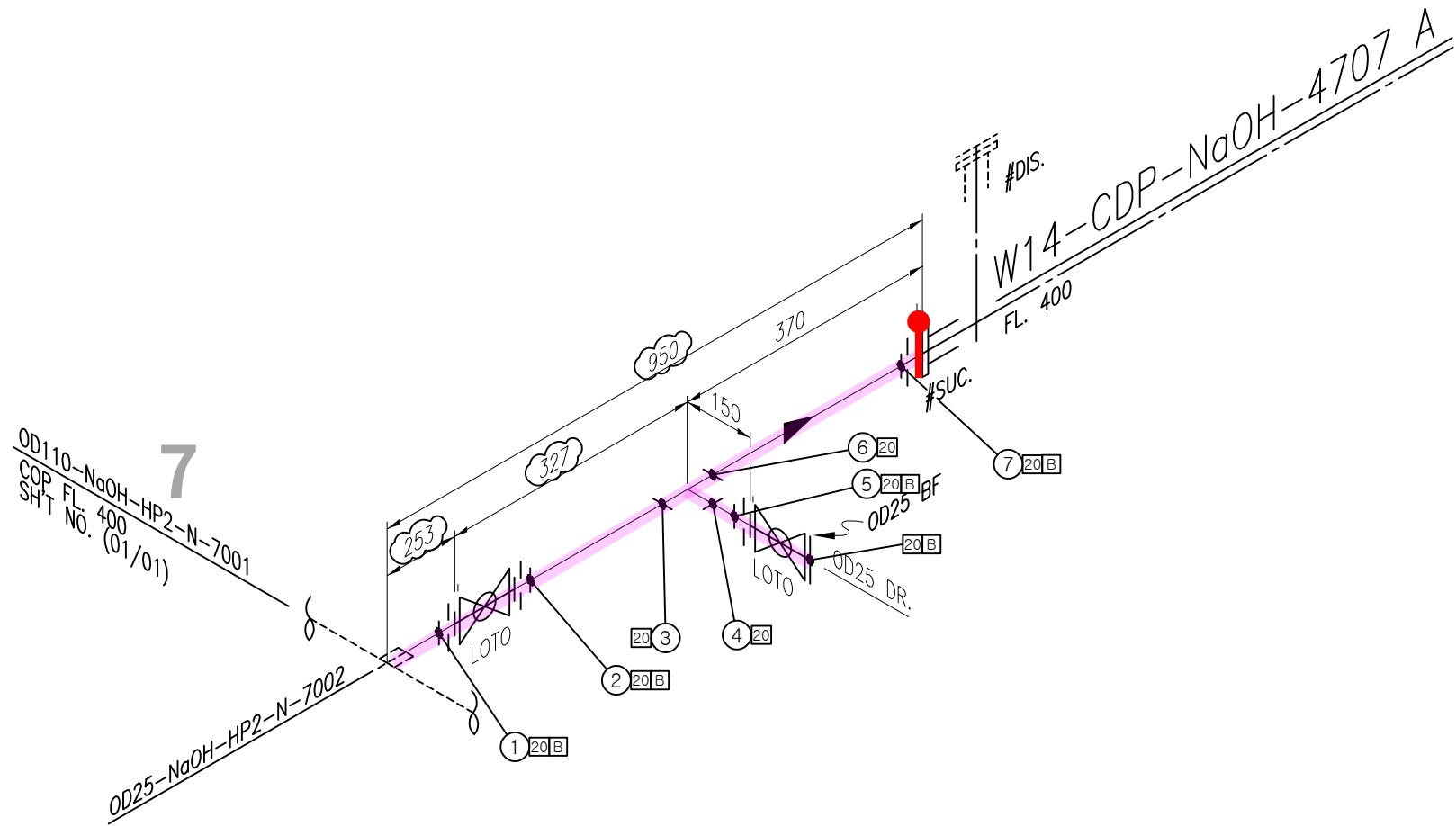
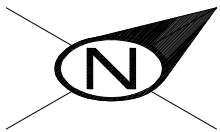
대외비

KEY PLAN (4F)

OTE
혼전 시 CLOSE 되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM / CHECK BALL
3. 악물종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
5. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAINAGE 인단의 TRENCH로 흡수되어, DRAIN HEADER로 이용된다.
6. PUMP VENDOR 공급.
7. 가성소다 LINE에 위치한 VALVE에 LOTO 적용.
8. INTERLOCK NO. I-4653
(W13-CTP-NaOH-4673 AUTO STOP)
: REMOTE STATUS에서 W14-T-NaOH-4607 A/B LEVEL (U-W6065 A/B) HIGH HIGH STOP.
9. INTERLOCK NO. I-4616
(W14-CDP-NaOH-4645 A/B/C,
W14-CDP-NaOH-4646 A/B/C,
W14-CDP-NaOH-4647 A/B/C,
W14-CDP-NaOH-4648 A/B/C,
W14-CDP-NaOH-4649 A/B/C AUTO STOP
: REMOTE STATUS에서 W14-T-NaOH-4607 A/B LEVEL (LI-W6066 A/B) LOW LOW STOP.
ALL SPARE PUMPS는 pH SET VALUE HIGH STOP.
10. N-STRIPPER PACKAGE TRAIN A/B 중 선택연전.
11. TANK의 역만 레벨 경계치는 VENDOR에서 공급한다.
12. 모든 가성소다 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
가성소다 환부 AREA에는 비산방지막을 설치한다.
14. OPERATING CAPACITY : 0.5 LPM
OPERATING PRESSURE : 0.59 MPa

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 EV.	실명 DESCRIPTION	날짜 DATE
DRAWN BY 이인학		
검토 한진희		
CHECKED BY 박동명		
승인		
APPROVED BY		
제작사		
SK 에코플랜트		
공사명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
NaOH STORAGE TANK & FEED PUMP (1/2)		
속척		
SCALE		
NONE		
날짜		
DATE		
2025. 08. 14		
PROJECT NO.		22330D
도면번호		CJ-M15W14-PR-A20-000154
SHEET NO.		2A
REV.		A





<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	1.OM	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD25	1EA	TEE , HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	4 EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
FB	OD25	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STSS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	4 EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD25	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VBLO	OD25	2 EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
SH	OD25	5EA	SHIELD

CLIENT



1

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
NaOH	22330D
LINE NO.	SH'T NO. REV.
OD25-NaOH-HP2-N-7002	01 / 01 0

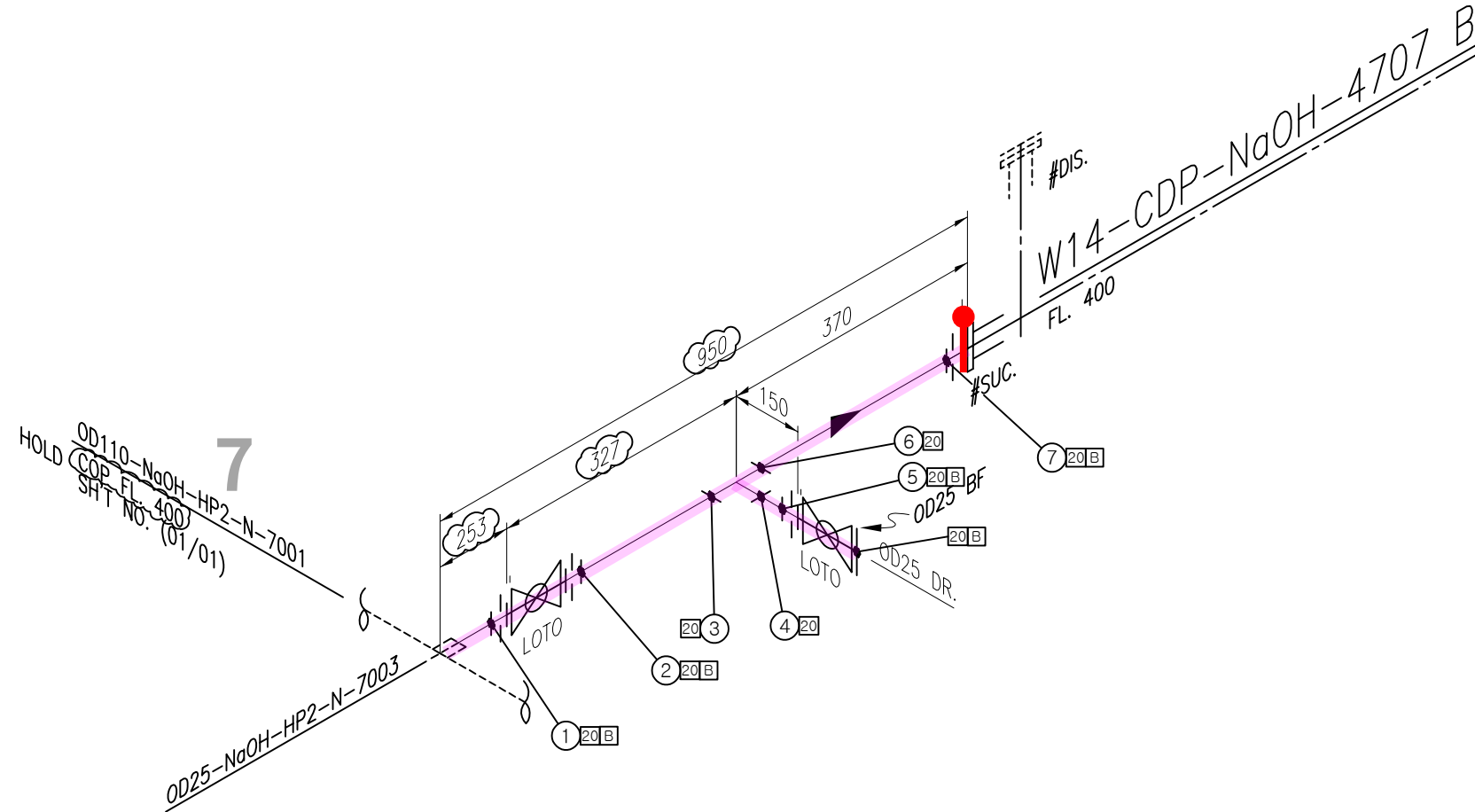
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	1	60	UNIT	4FA								
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040012-A	△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	JOB NO.
						REF. P & ID	CJ-M15W14-PR-A20-000318								
														LINE NO.	SH'T NO. REV.
														OD25-NaOH-HP2-N-7002	01 / 01 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	1.0M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD25	1EA	TEE , HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	4 EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
FB	OD25	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND–STS304
			/ LINDER–HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	4 EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD25	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VBLO	OD25	2 EA	BALL VALVE, KS 10K, FF FLG'D, BODY–PP /
			TRIM–PP / SEAT–PTFE, FB, LEVER + LOTO
SH	OD25	5EA	SHIELD



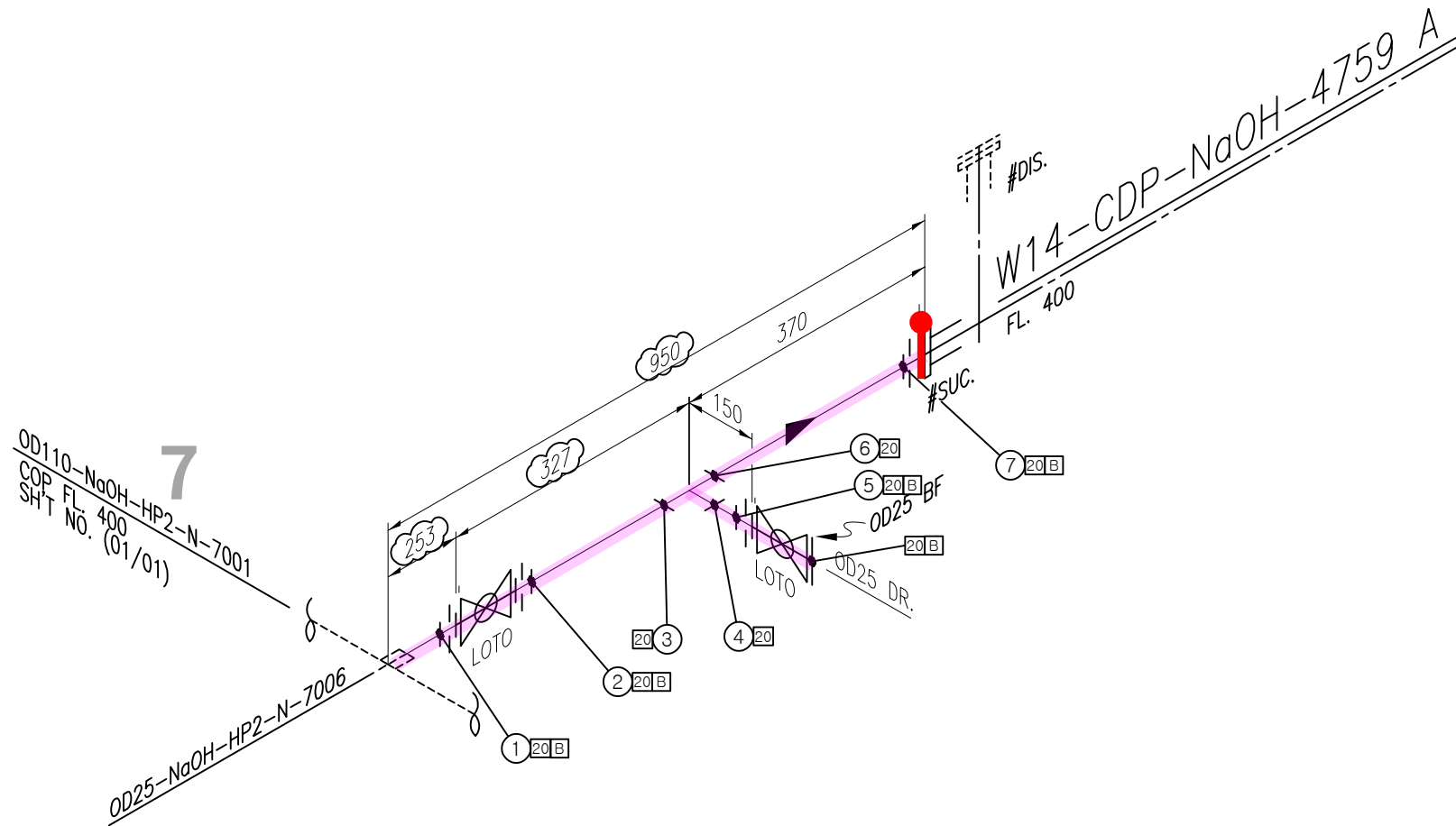
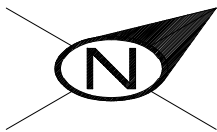
<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	60	UNIT	4FA							M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG.							COMMODITY	JOB NO.		
						CJ-M15W14-PI-A30-040012-A							NaOH	22330D		
						REF. P & ID			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000318		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	OD25-NaOH-HP2-N-7003	01 / 01	0

[illegible]



<NOTE>
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배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	1.0M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD25	1EA	TEE , HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	4 EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
FB	OD25	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	4 EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD25	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VBLO	OD25	2 EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
SH	OD25	5EA	SHIELD

CLIENT



3

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
NaOH	22330D
LINE NO.	SH'T NO. REV.
OD25-NaOH-HP2-N-7006	01 / 01 0

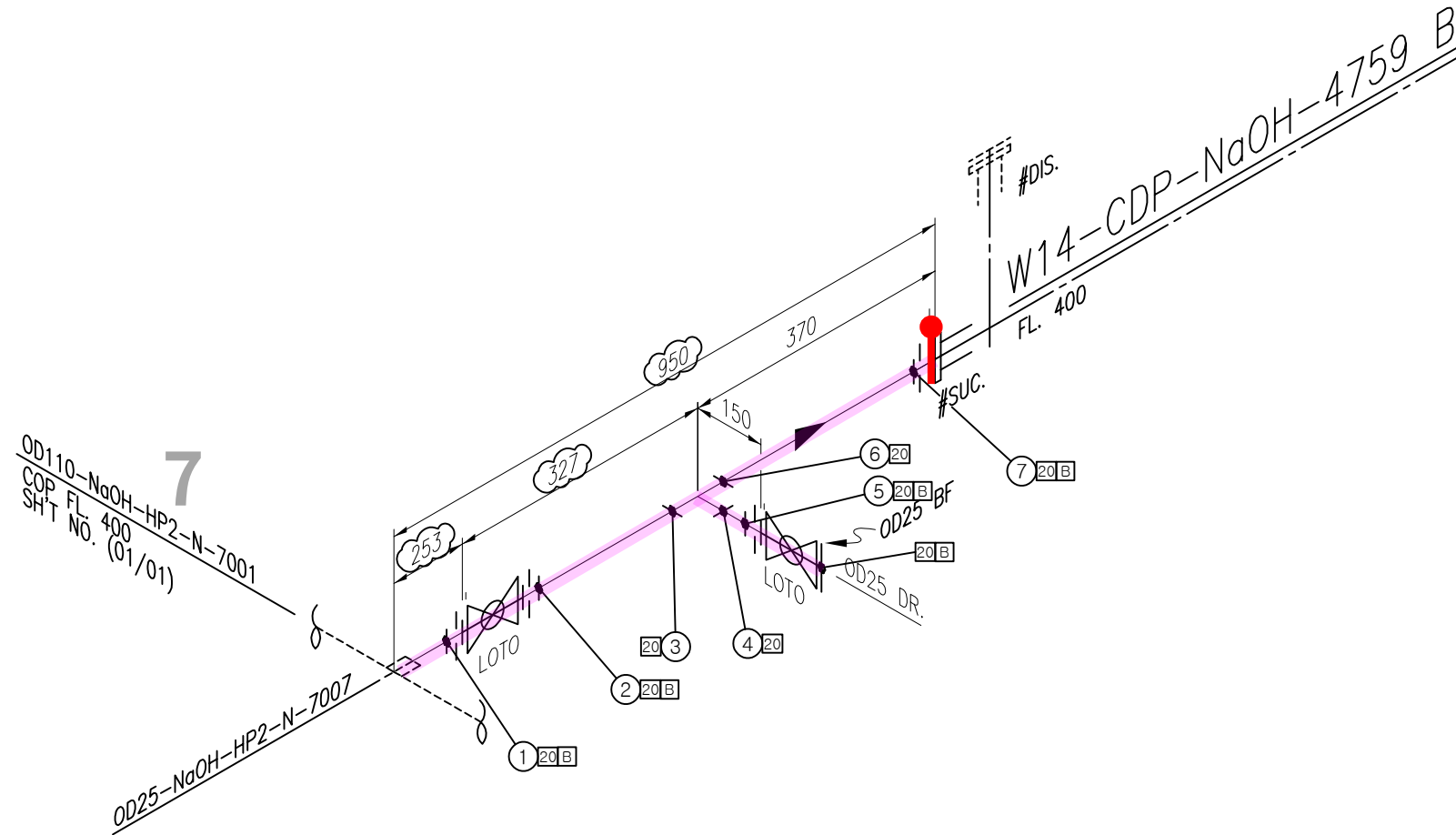
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	1	60	UNIT	4FA								
			OPERATING	ATM	AMB	REF. PLAN DWG.									
						CJ-M15W14-PI-A30-040012-A									
						REF. P & ID									
						CJ-M15W14-PR-A20-000318									

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	1.0M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD25	1EA	TEE , HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	4 EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
FB	OD25	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STSS304
			/ LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	4 EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD25	5SET	MACHINE BOLT & NYLON NUTS .STS 304 / W WASHERS
VBLO	OD25	2 EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP /
			TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
SH	OD25	5EA	SHIELD



<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2L0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDL0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040012-A							COMMODITY	NaOH	JOB NO.
						REF. P & ID	CJ-M15W14-PR-A20-000318							LINE NO.	OD25-NaOH-HP2-N-7007	SH'T NO.
										25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		
									REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		01 / 01

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

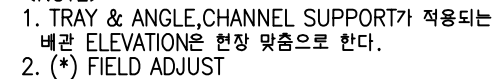


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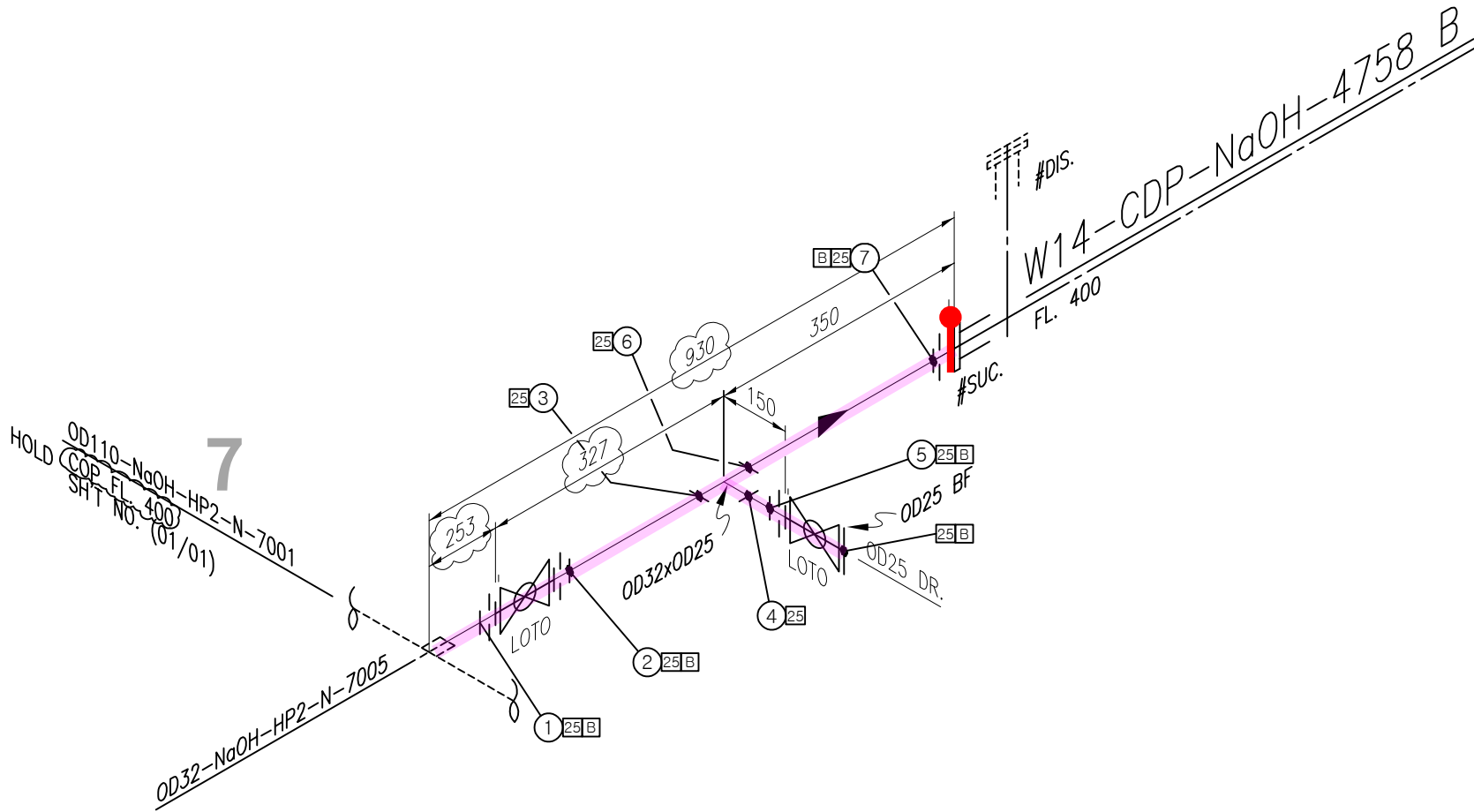
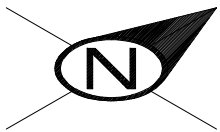


OD32-NaOH-HP2-N-7004

EV.



REV



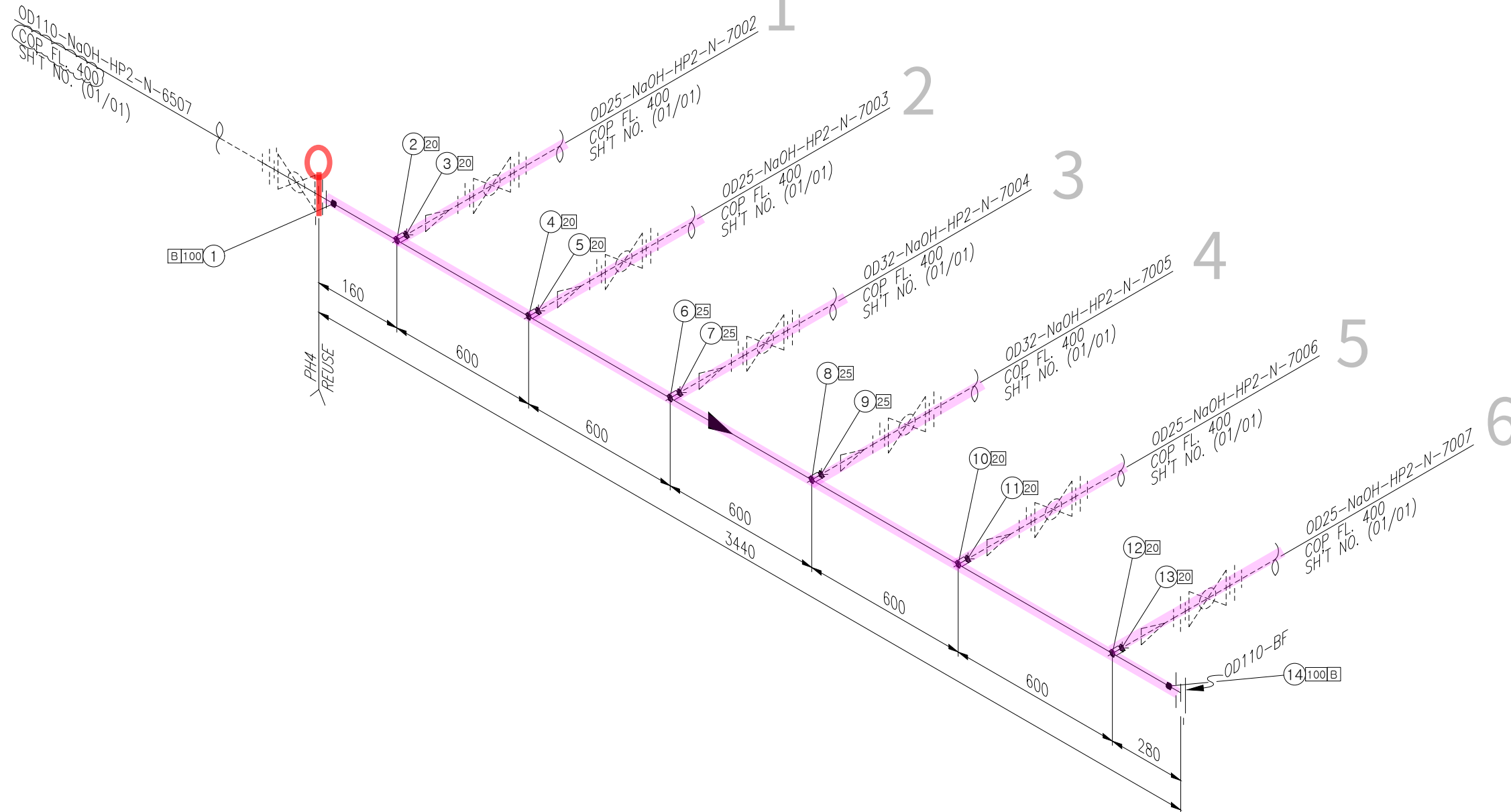
<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR		PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT	4FA									M15X Ph4 Project			
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040012-A										COMMODITY	NaOH	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000318			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			LINE NO.	OD32-NaOH-HP2-N-7005	SH'T NO. 01 / 01	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D						



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD110	3.5M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
SD	OD110xOD32	2EA	SADDLE, HDPE PE100, PE, BRANCH-SOCKET FUSION,
SD	OD110xOD25	4EA	SADDLE, HDPE PE100, PE, BRANCH-SOCKET FUSION,
F	OD110	2EA	FLG & SADDLE / STUB END SET, KS 10K, FF,
FB	OD110	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST3304
G2	OD110	1EA	GLASS REINFORCED HDPE (RFR HDPE COMBIBOND) 3.0MM THK
B1	OD110	1SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	OD110	1EA	SHIELD



<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR	PROJECT NAME		
	THICK.	-	DESIGN	1	60	UNIT 4FA							M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040012-A						COMMODITY	JOB NO.		
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOH	22330D	
						REF. P & ID CJ-M15W14-PR-A20-000318	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
												0D110-NaOH-HP2-N-7001	01 / 01	0	